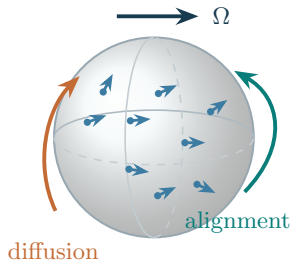


I Angular relaxation

Noise and aligning torque



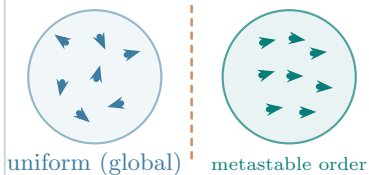
Fixed J : $r = k(J)$
angular rate $\tau_{\text{noise}} \lambda_d(r) > 0$

II Feedback and basins

Same density, different initial states

$$J \longrightarrow r = k(J) \longrightarrow \rho c_d(r) = J$$

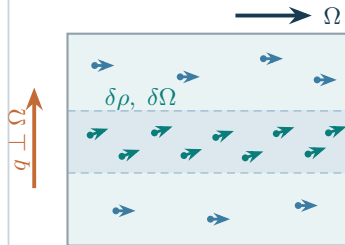
schematic basin boundary



$\rho = 1.90$: homogeneous PDE example

III Spatial SOH coupling

Density-direction coupling



$$\Theta < 0 \implies v_{\pm} \notin \mathbb{R}$$

first-order SOH; transverse waves

$K_2 > 0$: diffusive dispersion